Example: Action of malonate on succinate dehydrogenase.	Example: Action of cyanide on cytochrome oxidase.
Applied in drug action	Applied in toxicological studies.

9.3.2 Allosteric inhibition: (Greek allo = 'other'; stereos = 'space' or 'site')

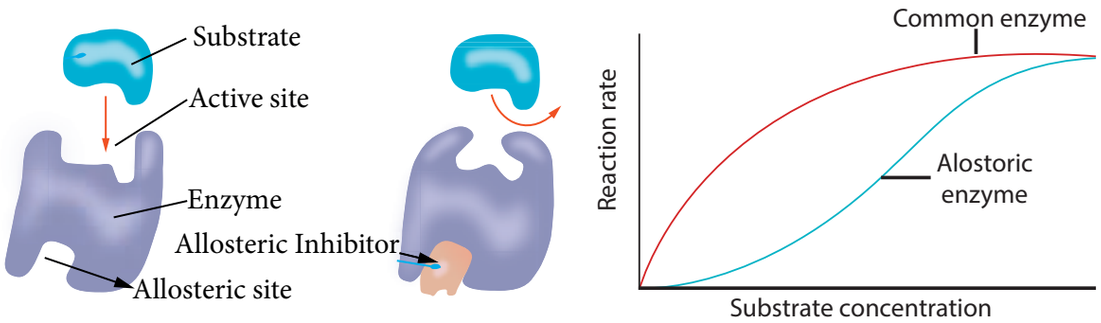


Figure 9.13 Allosteric inhibition

The allosteric inhibitor binds to an enzyme at a site other than the active site (Allosteric site). This leads to a conformational change which reduces the affinity of the enzyme for substrate (Figure 9.13). Allosteric enzymes do not exhibit Michaelis Menten kinetics. A graph of v_0 versus [S] gives a sigmoidal curve.

9.3.3. Irrversible enzyme inhibition :

When the inhibitor binds tightly to the enzyme either covalently or noncovalently and dissociates very slowly from the enzyme - inhibitor complex, it is known as irreversible inhibition (Figure 9.14)

Example :

1) Action of nerve gases on acetylcholine esterase.
2) Action of iodoacetamide on enzymes containing –SH group by modifying cysteine.

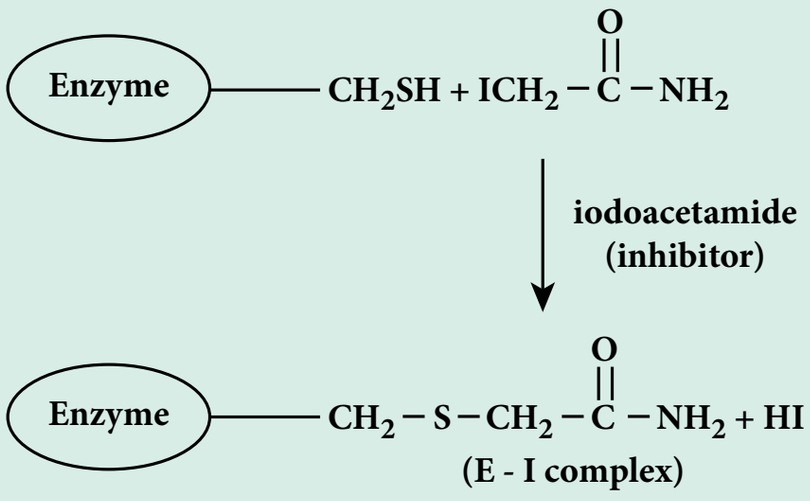


Figure 9.14 Irreversible Inhibition

XII U9 Enzyme Kinetics.indd 221 4/8/2020 3:27:16 PM